SOCIAL MEDIA ANALYTICS

THE CASE FERRAGNI-BALOCCO

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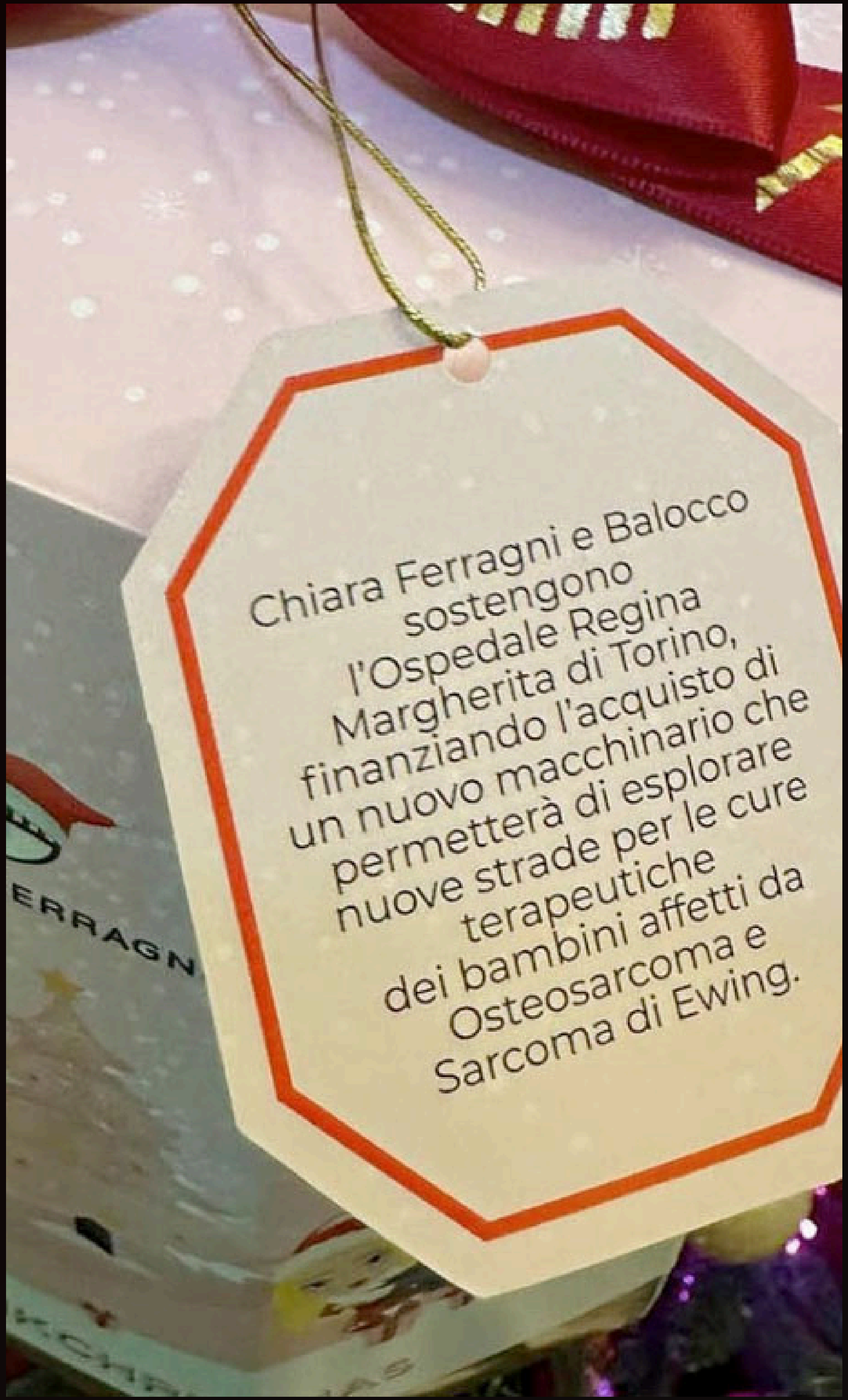
IN DECEMBER 2022, CHIARA FERRAGNI PROMOTED BALOCCO'S “PINK CHRISTMAS” PANDORO, SUGGESTING THAT PURCHASING IT WOULD SUPPORT THE REGINA MARGHERITA HOSPITAL IN TURIN.

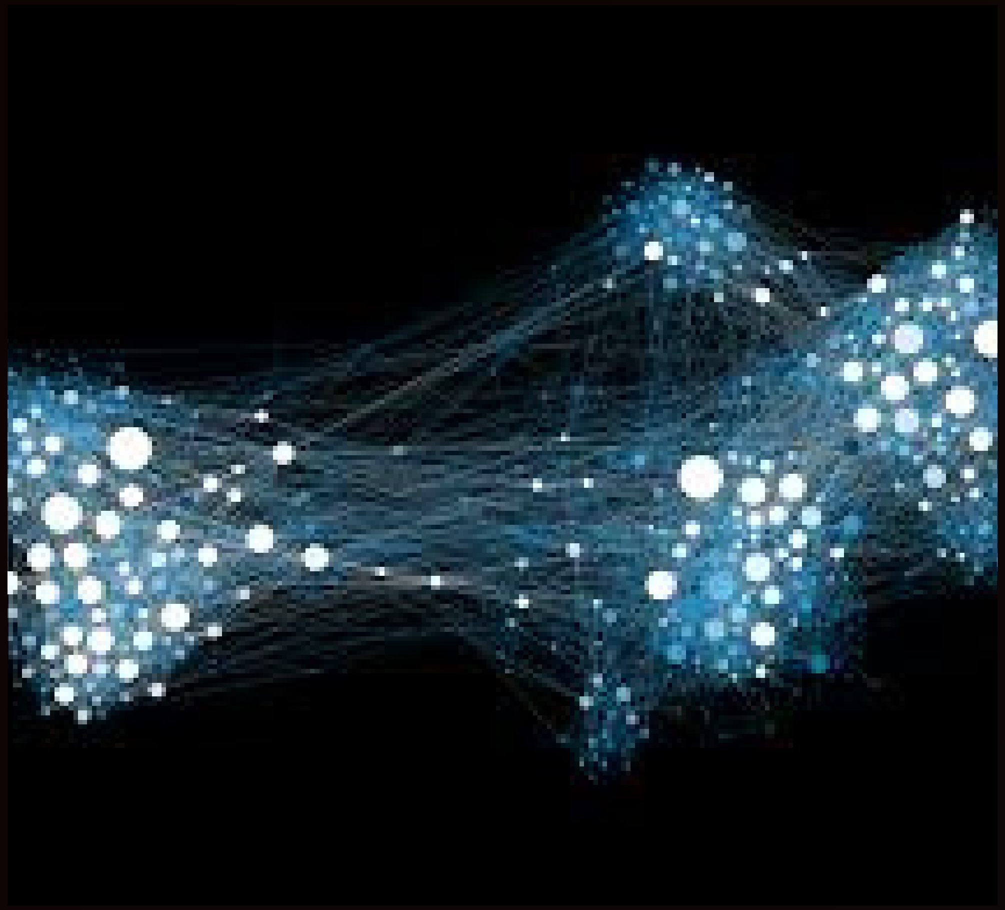
AN INVESTIGATION BY THE ITALIAN ANTITRUST AUTHORITY (AGCM) REVEALED THAT BALOCCO HAD ALREADY MADE A €50,000 DONATION MONTHS EARLIER, REGARDLESS OF SALES.

IN DECEMBER 2023, CHIARA FERRAGNI'S COMPANIES WERE FINED OVER €1 MILLION FOR UNFAIR COMMERCIAL PRACTICES, FOLLOWED BY THE OPENING OF AN INVESTIGATION FOR AGGRAVATED FRAUD



ON JANUARY 14, 2026, THE COURT OF MILAN DISMISSED THE CASE, ACQUITTING CHIARA FERRAGNI OF CHARGES OF AGGRAVATED FRAUD. THE RULING ESTABLISHED THAT THERE WERE NO AGGRAVATING CIRCUMSTANCES, CONFIRMING THAT THE CASE, DESPITE HAVING HAD A HUGE IMPACT ON HER REPUTATION, DID NOT MEET THE REQUIREMENTS FOR A CRIMINAL TRIAL.

OVERVIEW



DATA COLLECTION

- **Data extraction from Reddit**

SOCIAL NETWORK ANALYSIS

- **Reply Network**
- **Community Detection**

SOCIAL CONTENT ANALYSIS

- **Sentiment ed Emotion Analysis**

DATA COLLECTION

REDDIT SUBSEARCH

USING THE PRAW LIBRARY (PYTHON REDDIT API WRAPPER) FOR TARGETED EXTRACTION OF POSTS AND COMMENTS FROM THE SUBREDDITS R/ITALY, R/ITALIA, AND R/FAUXMOI

GLOBAL SEARCH VIA KEYWORD

POSTS WERE SELECTED USING SPECIFIC KEYWORDS RELATED TO THE FERRAGNI-BALOCCO CASE (E.G., “FERRAGNI,” “PANDORO”). THE DATASET WAS THEN REFINED BY FILTERING ONLY THREADS WITH AT LEAST FIVE COMMENTS IN ORDER TO ANALYZE ONLY SIGNIFICANT INTERACTIONS.

REPLY NETWORK

COMPLETE NETWORK

NODES	1349
EDGES	2165
ASSORTATIVITY	-0.07
AVG. DEGREE	1.6
DENSITY	0.0011

GIANT COMPONENT

NODES	1156
EDGES	1,554
ASSORTATIVITY	-0.106
AVG. S.P.	6.11
DENSITY	0.002

TOP USERS

Closenness & Betweenness

- MIRIESTE
- PURE-CONTACT7322
- V_FRLN

COMMUNITY DETECTION

Algorithm	Modularity	N°Communities
LOUVAIN	**0.7828**	**27**

COMMUNITY 2

Information of the case

Balocco, bambino, panettone, campagna, artifizio, multa, pandoro

COMMUNITY 0

Institutional & Legal

Manipolazione, mercato, cura, procedimento giudiziario, reato, pubblicità, giudice, reputazionale

COMMUNITY 1

Punitive Discourse

Carcere, furto, pena, condizionale, condanna, fedina

Topic Modeling via LDA

•**Goal**: Uncover latent thematic structures within specific user communities.

•**Approach**: Local LDA (applied at the community level) instead of Global LDA to capture granular nuances. Instead of a "One-Size-Fits-All" approach, LDA was iterated across the Top 10 communities.

PRE-PROCESSING

•**Noise Neutralization** (Regex): Stripped URLs, mentions, and hashtags.

•**Linguistic Filtering** (POS Tagging): Restricted vocabulary to *Nouns* and *Adjectives* .

•**Lemmatization**: Reduced all Italian inflected forms to their dictionary base.

•**Domain-Specific Stopwords**: Augmented standard lists with a custom lexicon to remove high-frequency entities,social jargon, and profanity.

MODEL CONFIGURATION

Implemented via the Gensim library with the following parameters:

•**Dictionary Pruning:** Removed terms appearing in < 5 documents or > 50% of the corpus to eliminate outliers and generic terms.

•**Topic Cardinality** (k=3): Selected for superior Coherence Scores and optimal interpretability, ensuring well-defined thematic boundaries.

EVALUATION AND VALIDATION OF LDA

To ensure scientific rigor, the results were validated through two lenses.

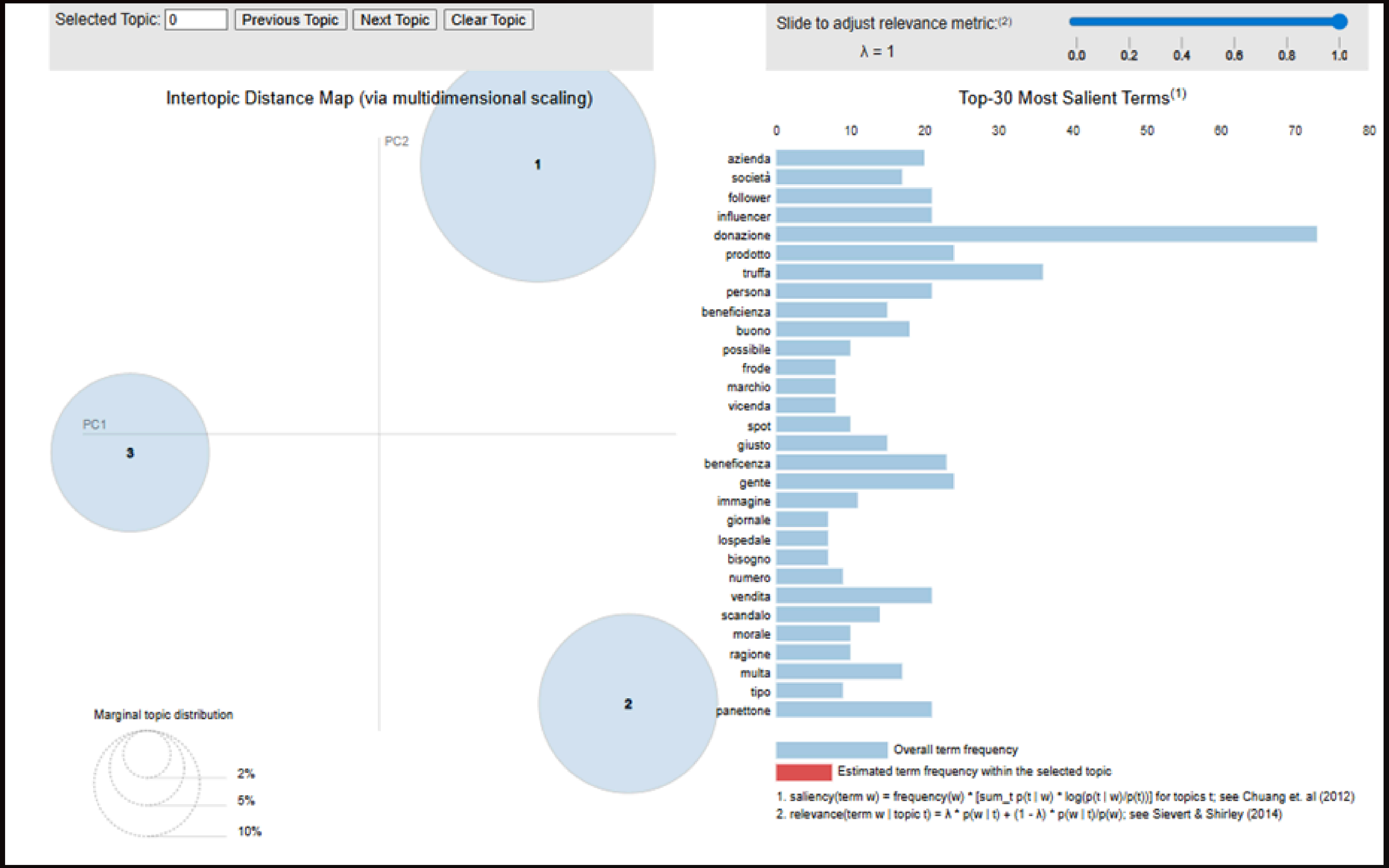
Quantitative (Coherence Score)

The model achieved a score of ***0.43.*** In the fragmented context of social media comments, this indicates a strong level of thematic alignment and internal consistency.

Qualitative (pyLDAvis)

Visual inspection of the Inter-topic Distance Map confirmed:

- •High Saliency: Terms clearly distinguish one topic from another.
- •Zero Overlap: Sharp and neat boundaries between the three areas of discourse, proving the topics are conceptually unique.



pyLDAvis of Community 2

SENTIMENT ANALYSIS with VADER

1. Why VADER?

•Specifically tuned for social media contexts.

Nuance Detection: Unlike standard models, it accounts for:

- Punctuation (e.g., "!!!" increases intensity).
- Capitalization (e.g., "ANGRY" vs "angry").
- Emojis, which are prevalent in our dataset.

3. Classification Thresholds

Sentiment is determined by the Compound Score:

LABEL	THRESHOLDS COMPOUND SCORE
POSITIVE	>= 0.05
NEUTRAL	BETWEEN -0.05 AND 0.05
NEGATIVE	<= -0.05

2. "Context-Aware" Pre-processing

To maximize VADER's effectiveness, we maintained the text's emotional integrity:

- •Minimal Cleaning: Only URLs, HTML tags, and @mentions were removed.
- •Preservation: Capitalization and punctuation were kept to allow the detection of emotional intensity.
- •Translation: Analysis was applied to the translated English version of the corpus.

SENTIMENT ANALYSIS with AFINN

1.Why AFINN?

AFINN is a pure lexicon-matching model. It assigns a specific valence score (from -5 to +5) to individual words.

2. Methodology: Lexicon-Matching

- Valence Scoring: AFINN assigns numerical scores to individual words, ranging from -5 (Highly Negative) to +5 (Highly Positive).
- Normalization Strategy: To ensure fairness across comments of different lengths, we calculated a Normalized Score ($Score / Word Count$). This prevents longer comments from having artificially inflated sentiment weights.

3. Dictionary-Specific Pre-processing

Designed to maximize the "hit rate" of the AFINN dictionary:

- Lowercasing: Standardized text for case-insensitive matching.
- Punctuation Removal: Isolated individual tokens for cleaner lookup.
- Emoticon Support: Enabled *“emoticons=True”* to capture emotional intent from symbols/emojis.

AFINN RESULTS

POSTS

SENTIMENT LABEL	COUNT	PERCENTAGE
Negative	7	13.21%
Neutral	33	62.26%
Positive	13	24.53%

COMMENTS

SENTIMENT LABEL	COUNT	PERCENTAGE
Negative	2050	36.94%
Neutral	1606	28.94%
Positive	1894	34.13%

VADER RESULTS

POSTS

SENTIMENT LABEL	COUNT	PERCENTAGE
Negative	6	11.32%
Neutral	35	66.04%
Positive	12	22.64%

COMMENTS

SENTIMENT LABEL	COUNT	PERCENTAGE
Negative	1906	34.34%
Neutral	1382	24.90%
Positive	2262	40.76%

- **Sentiment Resilience:** Positive sentiment is remarkably high, nearly equal to or exceeding negative tones in both models.
- **Posts as Information Anchors:** Both models agree on a dominant Neutral classification (~62–66%).

PREPROCESSING & ARCHITECTURES

PREPROCESSING STRATEGY

- **Conservative Pipeline**: designed to retain affective signals.
- **Retained:** Emojis,
- punctuation, informal intensifiers.
- **Removed**: URLs and user mentions.
- **Truncation:** Capped at 512 tokens (Transformer limit).

SENTIMENT MODEL

- **Model:** twitter-xlm-roberta-base-sentiment.
- **Robustness**: specifically adapted to social media language and multilingualism.
- **Output**: three-class approach to explicitly handle neutrality.

EMOTION MODEL

- **Model**: feel-it-italian-emotion.
- **Logic**: categorical classification of a single dominant emotion.
- **Neutral**: treated as a baseline (absence of signal) and not as an emotional state.

GLOBAL DISTRIBUTION

POLARIZATION

70.0% OF THE CORPUS IS NEGATIVE. NEUTRALITY IS LOW (19.5%), INDICATING A DISCOURSE DRIVEN BY EMOTIONAL REACTION RATHER THAN INFORMATION EXCHANGE.

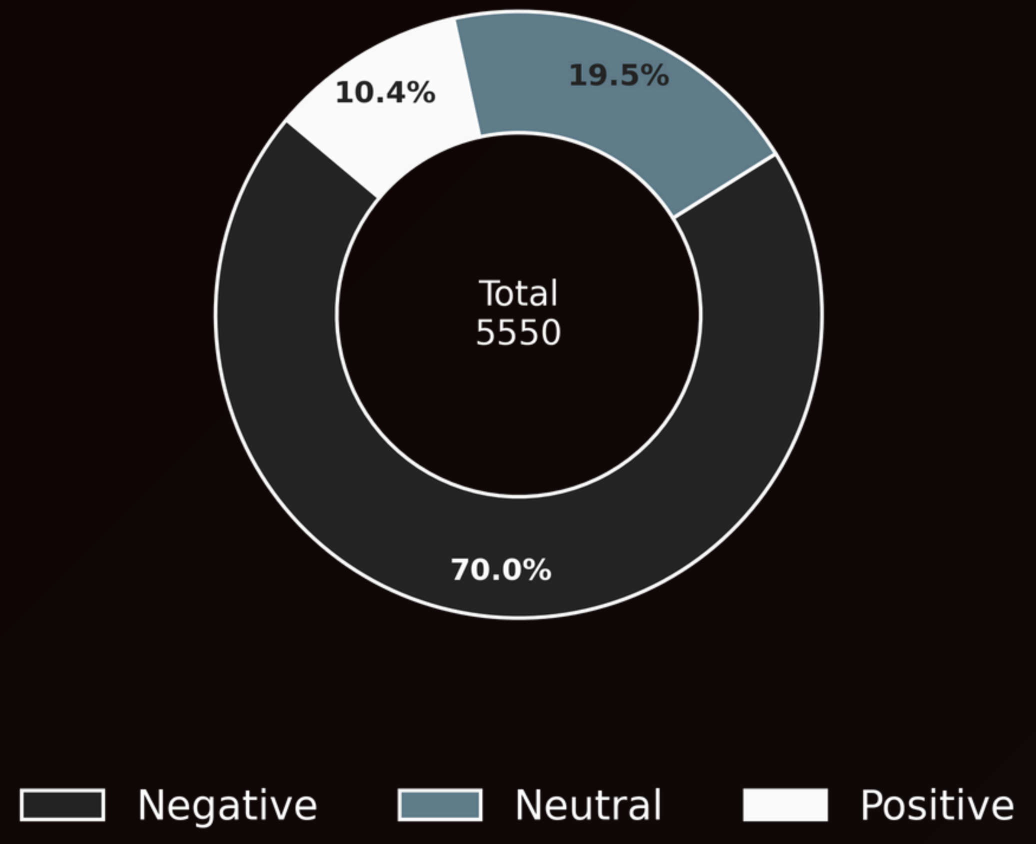
ANGER PREVALENCE

ANGER IS THE ABSOLUTE DOMINANT FORCE AT 62.2%. THE CONVERSATION IS FUELED BY MORAL OUTRAGE RATHER THAN PASSIVE DISAPPOINTMENT (SADNESS 15.4%).

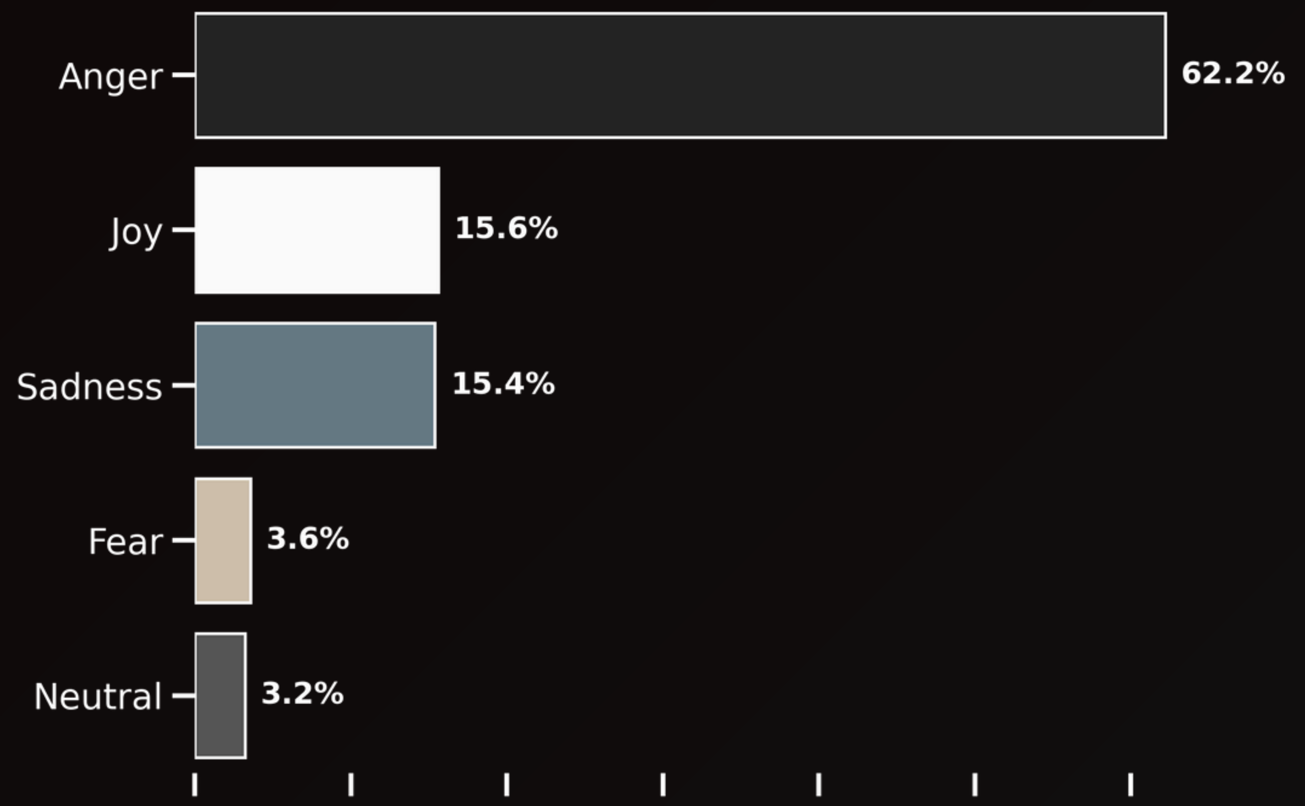
THE ANOMALY

15.6% JOY IMPLIES COUNTER-INTUITIVE POSITIVE AFFECT, REQUIRING QUALITATIVE DECONSTRUCTION.

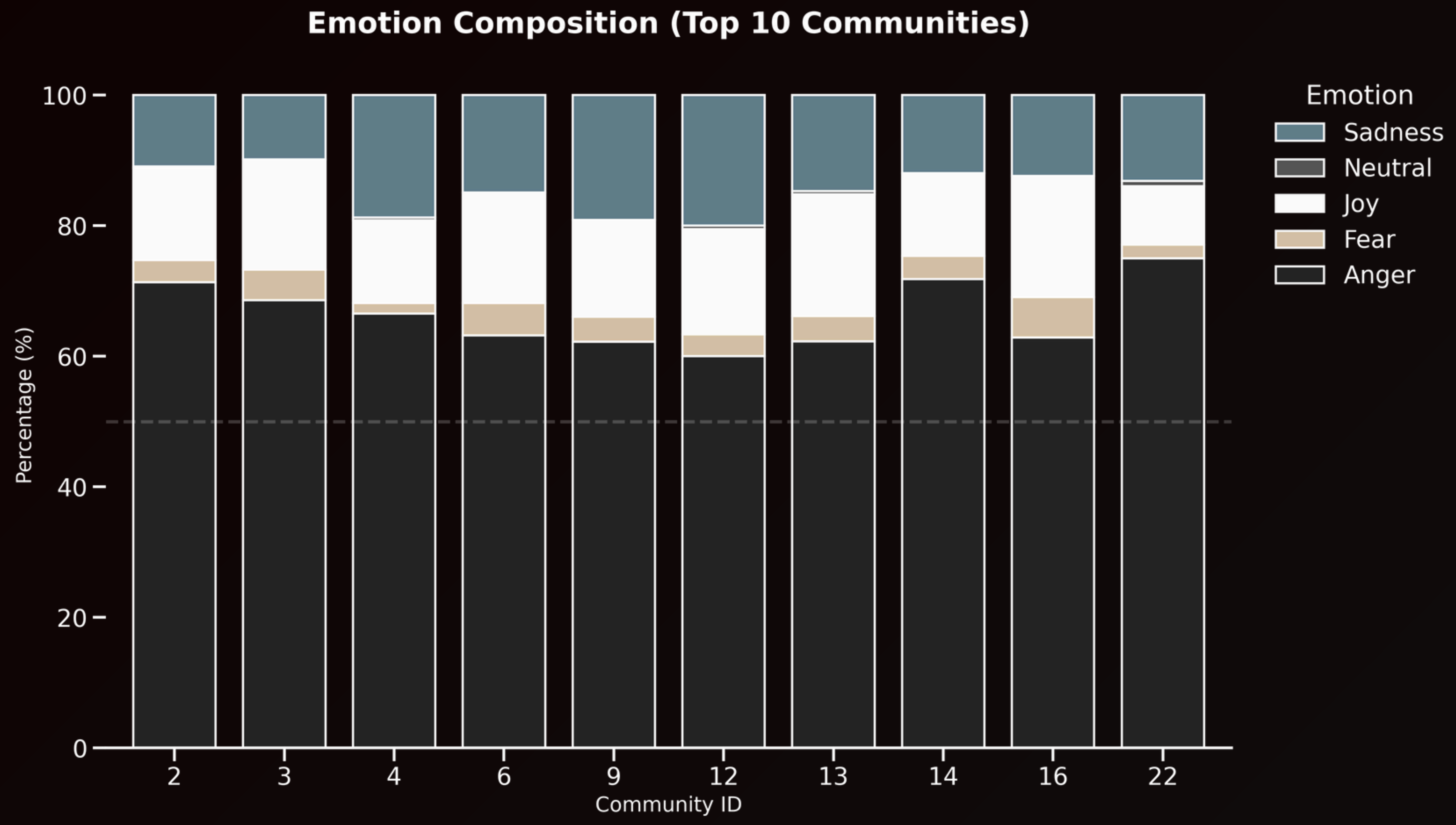
SENTIMENT POLARITY



DOMINANT EMOTIONS



COMMUNITY EMOTIONS ANALYSIS



KEY INSIGHTS

- **Uniform Criticality**: no community offers a 'safe harbor.' The network is uniformly critical.
- **Stylistic Variance**: while Anger is omnipresent, intensity varies. some communities are angrier; others show resignation (Sadness).
- Group differences: we can see that community 22 is the most dominated by anger (> 75%), while community 12 has an anger level of less than 60%.

DECONSTRUCTING THE JOY ANOMALY

MODEL CLASSIFICATION:

JOY

15,6 %

QUALITATIVE INSPECTION

PRAGMATIC REALITY

IRONY	“Il mercato prende cura di se stesso… sisi”
SARCASM	“Sempre meglio della Ferragni.”
SCHADENFREUDE	“Non vedo l’ora di leggere il libro che pubblicherà dal carcere.”

Conclusion: positive lexical markers often mask negative intent in scandal-driven discourse.

STATISTICAL VALIDATIONS AND CONCLUSIONS

STATISTICAL ROBUSTNESS

CHI-SQUARE TEST:

- **$\chi 2$** = 153.81.
- **p** = 0.001.
- **Result:** Rejects null hypothesis of independence.

PERMUTATION TEST:

- **p** = 0.003.
- **Effect Size:** Cramér's V = 0.104.
- **Small but reliable effect.**

STUDY LIMITATIONS

- **DOMAIN SPECIFICITY:** lack of human validation on Italian Reddit data.
- **IRONY BLINDNESS: c**urrent models struggle to identify genuine positive affect from mocking behavior.
- **Future Work:** implementation of specialized irony classifiers.

FINAL VERDICT

- The network shows **uniform criticism** expressed through distinct stylistic registers.
- While "Anger" is ubiquitous, the flavor of that anger varies significantly by community.
- Statistical tests confirm that these emotional patterns are not random, but **structural properties of the communities.**